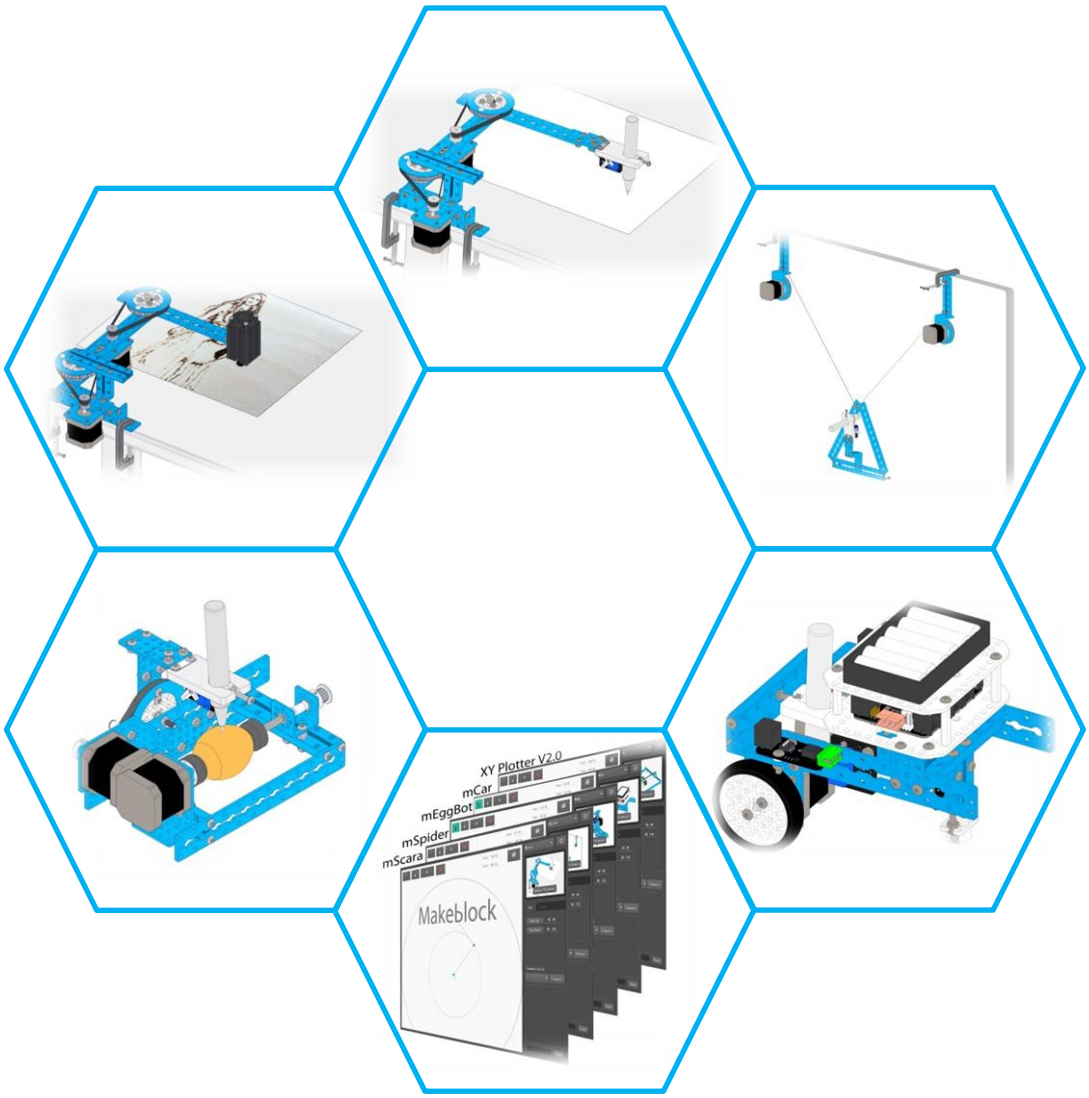
mDrawBot User Guide



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1. Before Get Start

1. Hardware:
   mDrawBot
   mDrawBot Mechanical Assemble Instruction

2. Software:
   a. Must have：
   mDraw
   Inksape

   b. Optional：
   Arduino IDE
   Makeblock mainboard Driver *(Note: If using Mac OS system, go to the website to download and install the driver. If using Windows system, mDraw software will automatically install the driver.)*

Tip 1

mDrawBot can be assembled into 4 configurations. For the accuracy of your robot, please strictly follow the assemble instruction, and carefully adjust your robot. Please pay attention to the position of axles, transmission parts, servo arm, and the levelness of robot.

Tip 2

Pictures in this article is only for reference. Specifications and software are subject to change without notice, please pay attention to our website http://www.makeblock.cc.

2. Part List (Standard Version)

Part list

1	Beam 0824-32	1		31	Plastic Caster Wheel	1
2	Beam 0824-96	2		32	12mm Stainless Steel Ball	1
3	Plate 0824-184	3		33	Rubber blanket	2
4	Bracket 3x3	3		34	Square Nut M4	1
5	Plate 3x6	3		35	Plastic Hinge A	3
6	Bracket U1	1		36	Plastic Hinge B	3
7	42BYG Stepper Motor Bracket B	2		37	Countersunk Screw M3x8	10
8	42BYG Stepper Motor	2		38	Headless Set Screw M3x5	10
9	Timing Pulley Slice 90T	4		39	Nylon Cap Nut M4	6
10	Plastic Timing Pulley 90T	2		40	Nut M4	40
11	Tire 90T B	2		41	Screws M4×8	26
12	2 Inch Gclamp	2		42	Screws M4×14	36
13	Pen Holder	1		43	Screws M4×16	16
14	Battery Holder for (6)AA	1		44	Screws M4×22	10
15	Makeblock Orion	1		45	Screws M4×35	2
16	Me Stepper motor driver	2		46	Nylon Stud M4×30	4
17	RJ25 Adapter	1		47	Plastic Rivet R4060	4
18	Wall Adapter Power Supply - 12VDC	1		48	Plastic Rivet R4120	4
19	Base Bracket V1.1	2		49	Plastic Ring 4x8x1mm	4
20	6P6C RJ25 cable-20cm	2		50	Plastic Ring 4x7x3mm	10
21	6P6C RJ25 cable-35cm	1		51	Plastic Ring 4x7x10mm	4
22	USB 2.0 A-Male to Micro B-Male Cable-3m	1		52	9g Micro Servo Pack	1
23	9g Servo Extension cord（3m)	1		53	Timing Belt 112MXL	2
24	Shaft Connector-4mm V2.3	2		54	Polyester thread	1
25	Timing Pulley18T	2		55	Nylon cable ties 2×100	10
26	Linear Motion Shaft D4x80mm	1		56	Rubber band	2
27	Spring 0.6×6×50	1		57	Cross &2.5mm HEX Screwdriver	1
28	Threaded Shaft 4x39mm	1		58	HEX Allen Key 1.5mm	2
29	Shaft Collar 4mm	2		59	Wrench 02	1
30	Flange Bearing 4x8x3mm	5		60	Instruction	1

3. Introduction

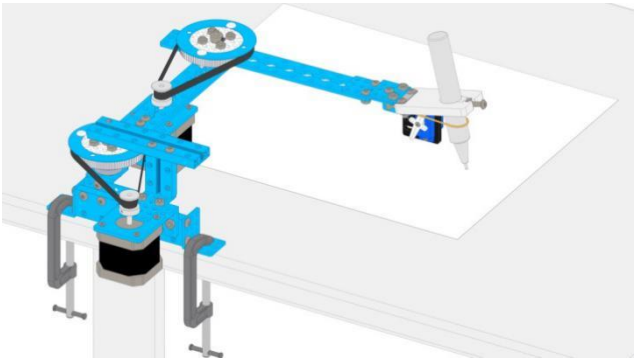
1. mDrawBot

mDrawBot is an advanced transformable drawing kit based on Makeblock platform. It can be assembled into 4 configurations which were named as mScara, mSpider, mEggBot and mCar. Besides the well-designed hardware, we designed a software for mDrawBot – mDraw.

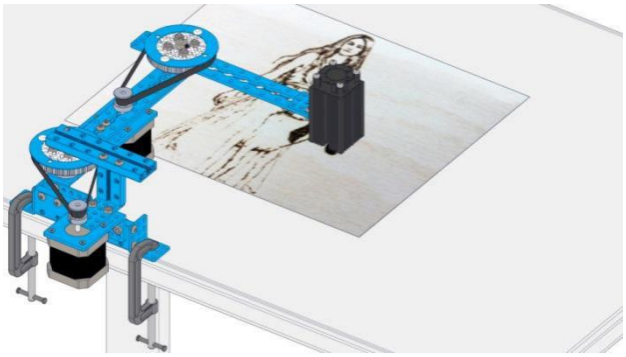
A. mScara

Scara stands for Selective Compliance Assembly Robot Arm. mScara is a drawing robot in cylindrical coordinate type which includes 2 structures.

a. Standard: the end of it is Pen lift mechanism.

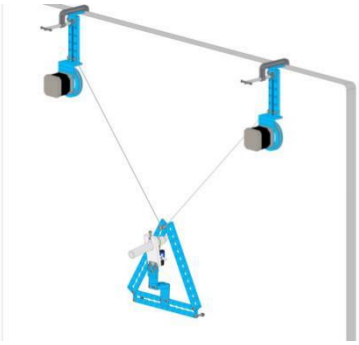


b. Advanced: the end of it is a 405nm/500mW blue-violet laser diode.



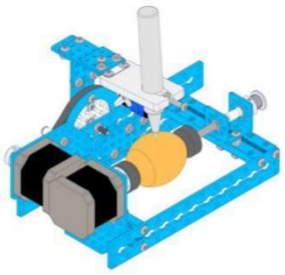
B. mSpider

mSpider is a wall drawing robot driven by 2 high-strength Dacron rope. This name comes from the spider-like structure.



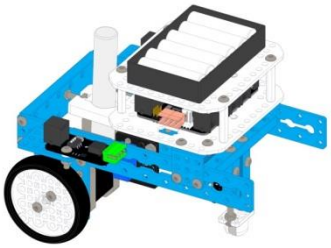
C. mEggBot

mEggBot is a drawing robot which is able to draw amazing pattern on the surface of eggs, column and spheroid.



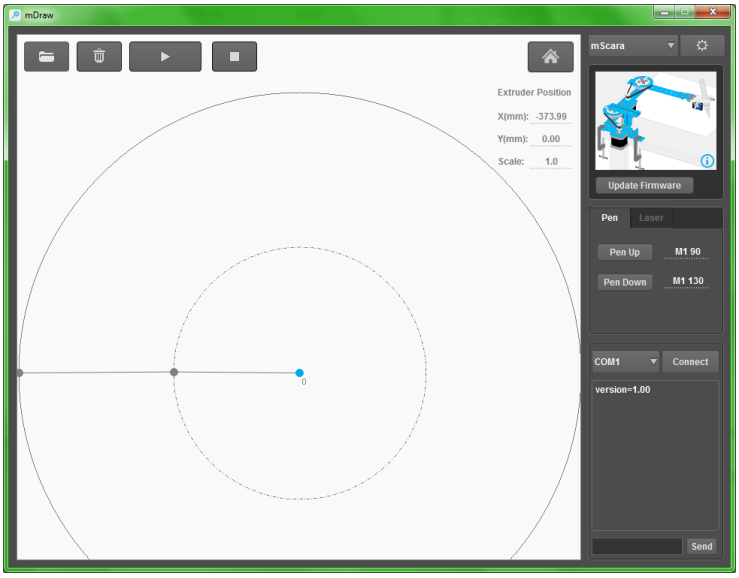
D. mCar

mCar is a 3-wheel drawing robot which can draw patterns in a large area on floor. Adding some electronics with DIY, and then it could upgrade to a self-balance car, a photographic car, a scout car and so on.



2. mDraw

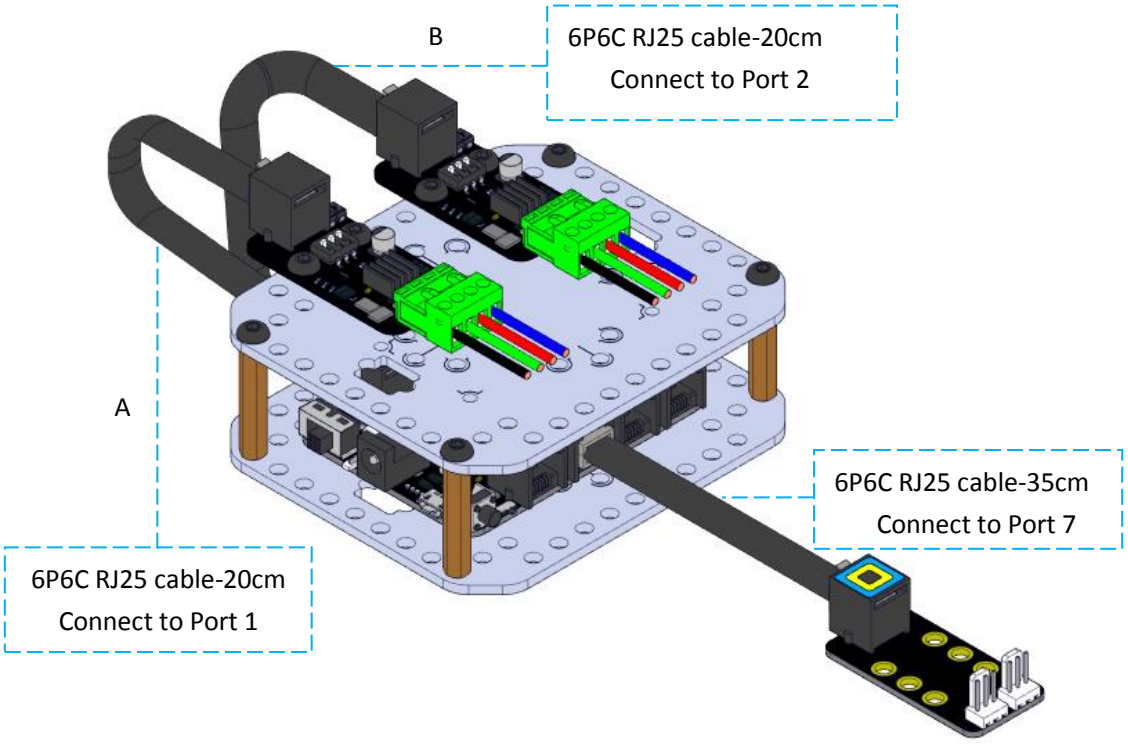
mDraw is a drawing software designed for mDrawBot. mDraw supports the 4 configurations of mDrawBot perfectly, and support the dual mode of Makeblock XY Plotter kit 2.0 as well.



mDrawBot is more than a robot integrated mechanic, electronics and software. mDrawBot is the entertaining robot that make art out of your hand.

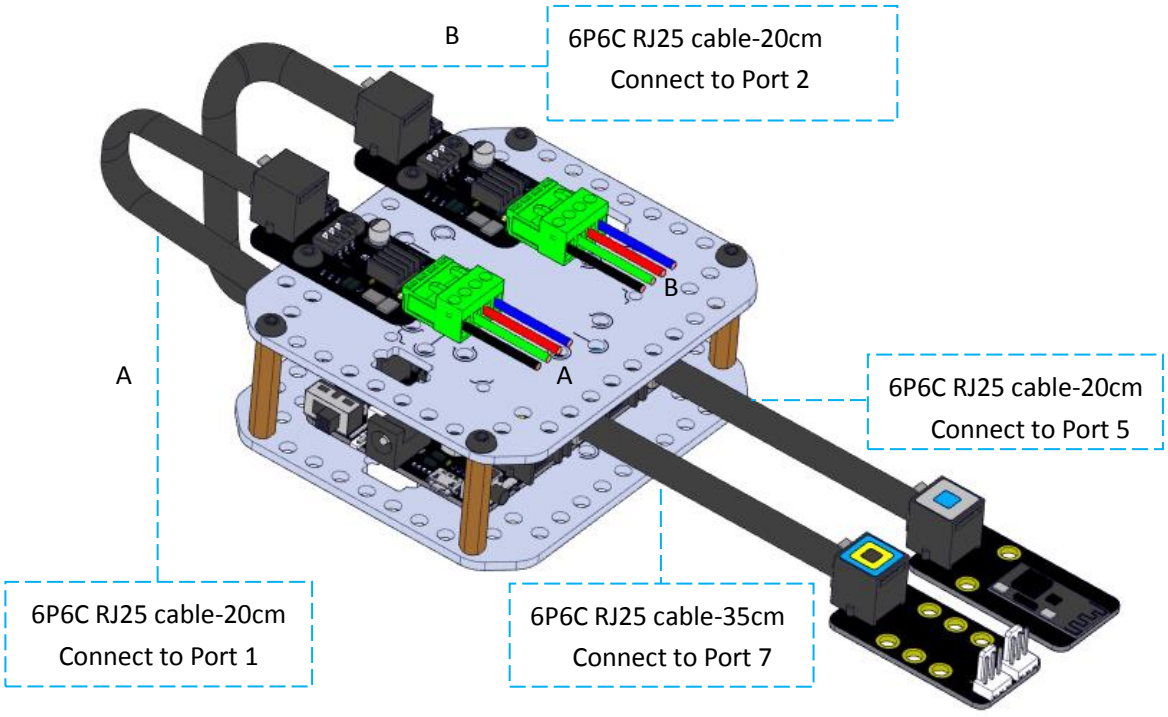
4. Recommended Layout of Wires

1. Standard Version



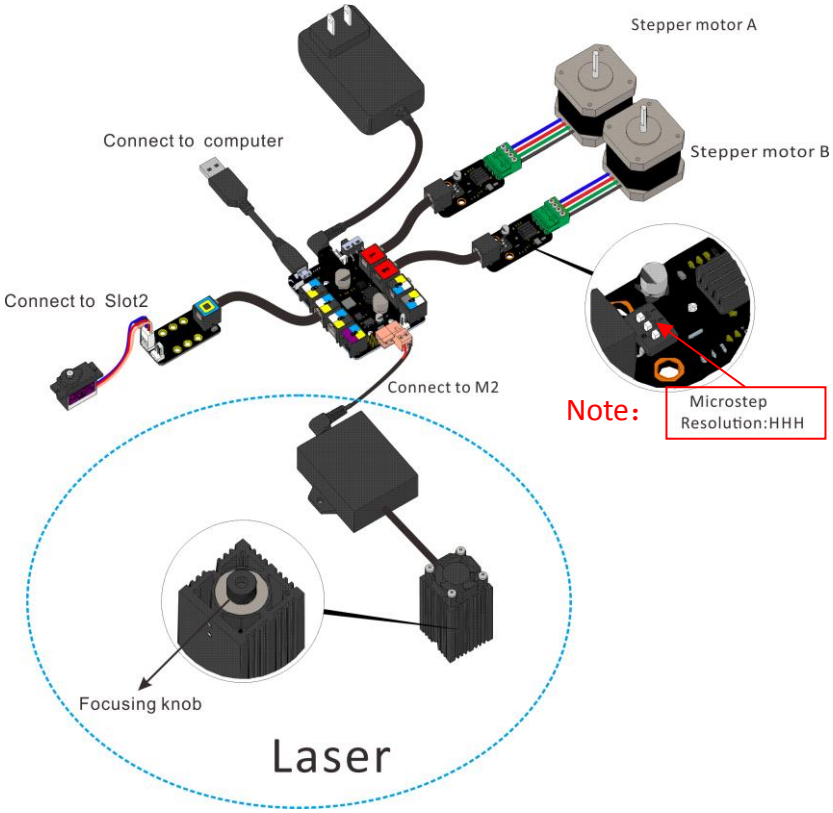
2. Bluetooth Version

(Attention: please connect Bluetooth module to Port_5 on Me Orion.)

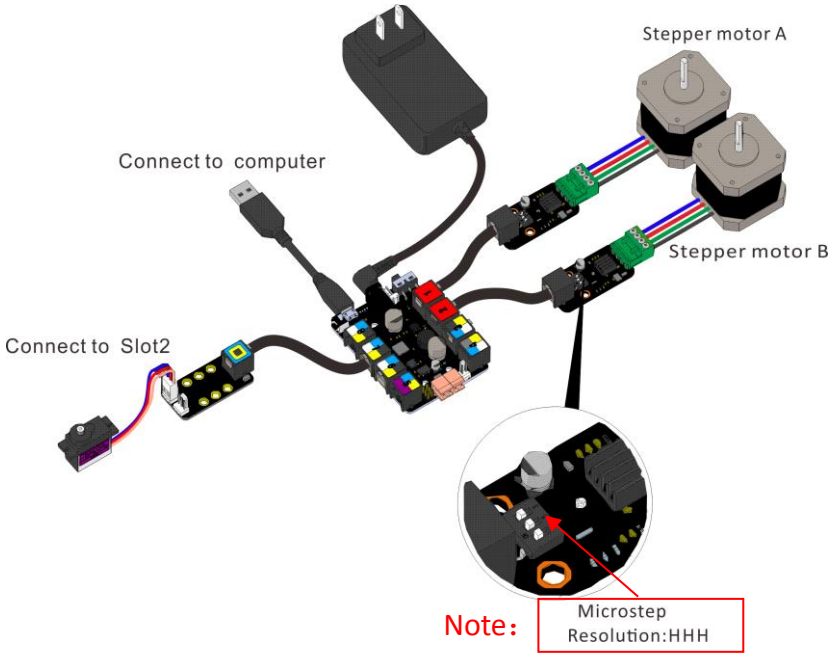


5. Wiring of mDrawBot

1. mScara



2. mSpider & mEggBot & mCar



After you finishing this step, congratulations! You can move forward to the next stage – using your mDrawbot to draw a picture. Here you go the introduction of software mDraw.

6. Software mDraw

1. Introduction of mDraw

mDraw is an cross-platform open-source software developed by Makeblock.

Compatibility: mDrawBot (mScara, mCar, mEggBot, mSpider), XY Plotter kit (servo mode, laser mode)

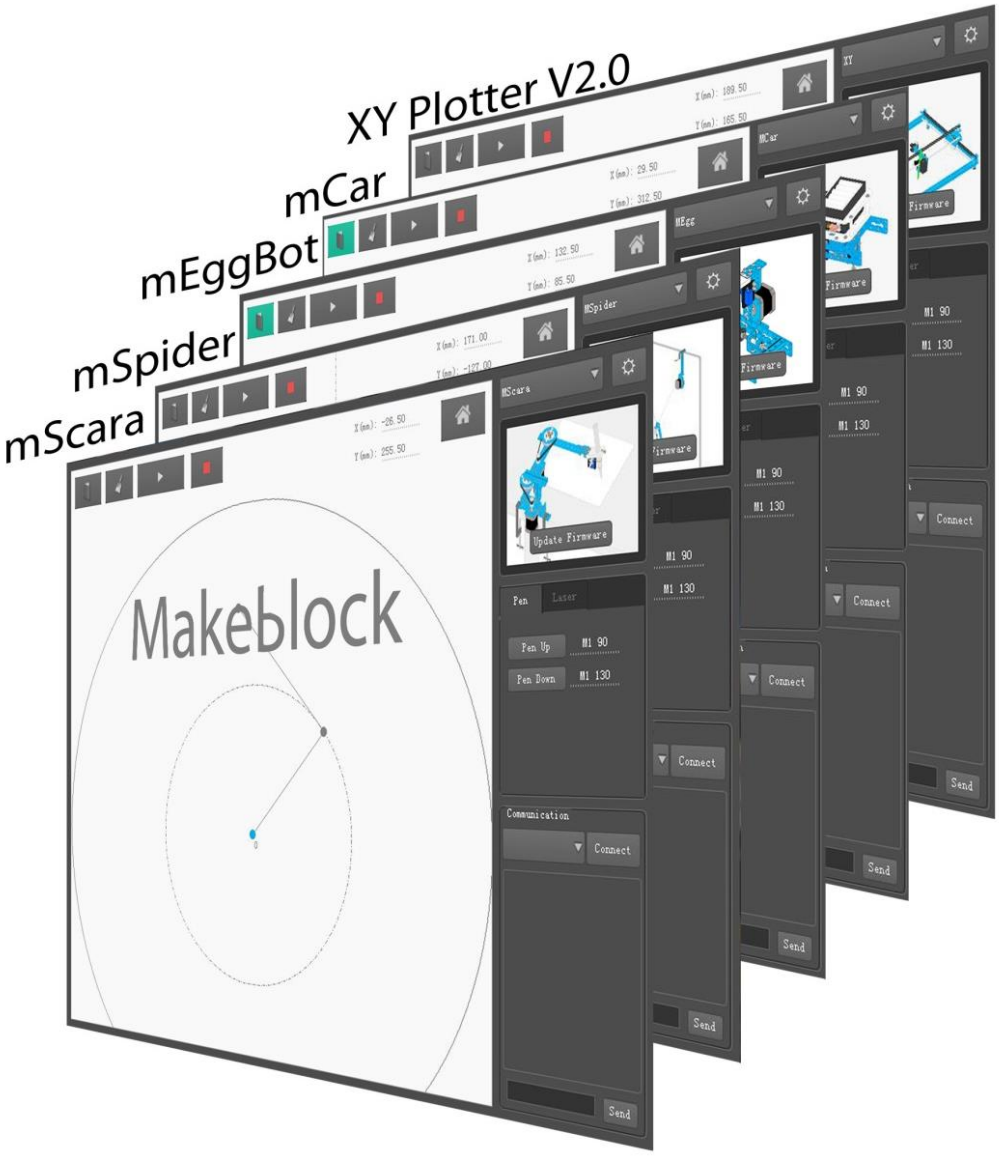
OS Environment: Windows, Mac, Linux

Supported File Type: *.svg, *.bmp (convert to *.svg)[1]

2. Installation

Download and open the installation pack of mDraw, follow the prompts to install mDraw.[2]

3. 5 in 1 Main UI



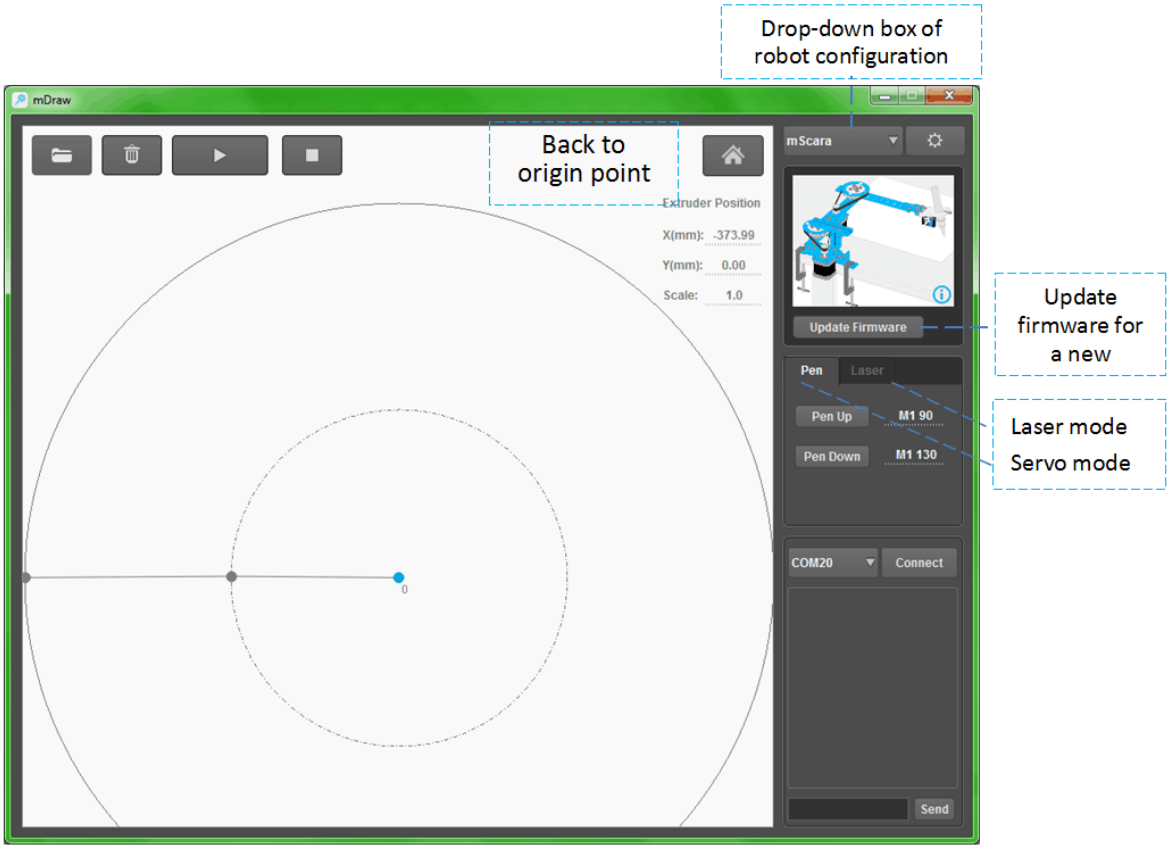
[1] All file must be named in English.

[2] The installation directory must be named in English.

4. Button Function in Main UI

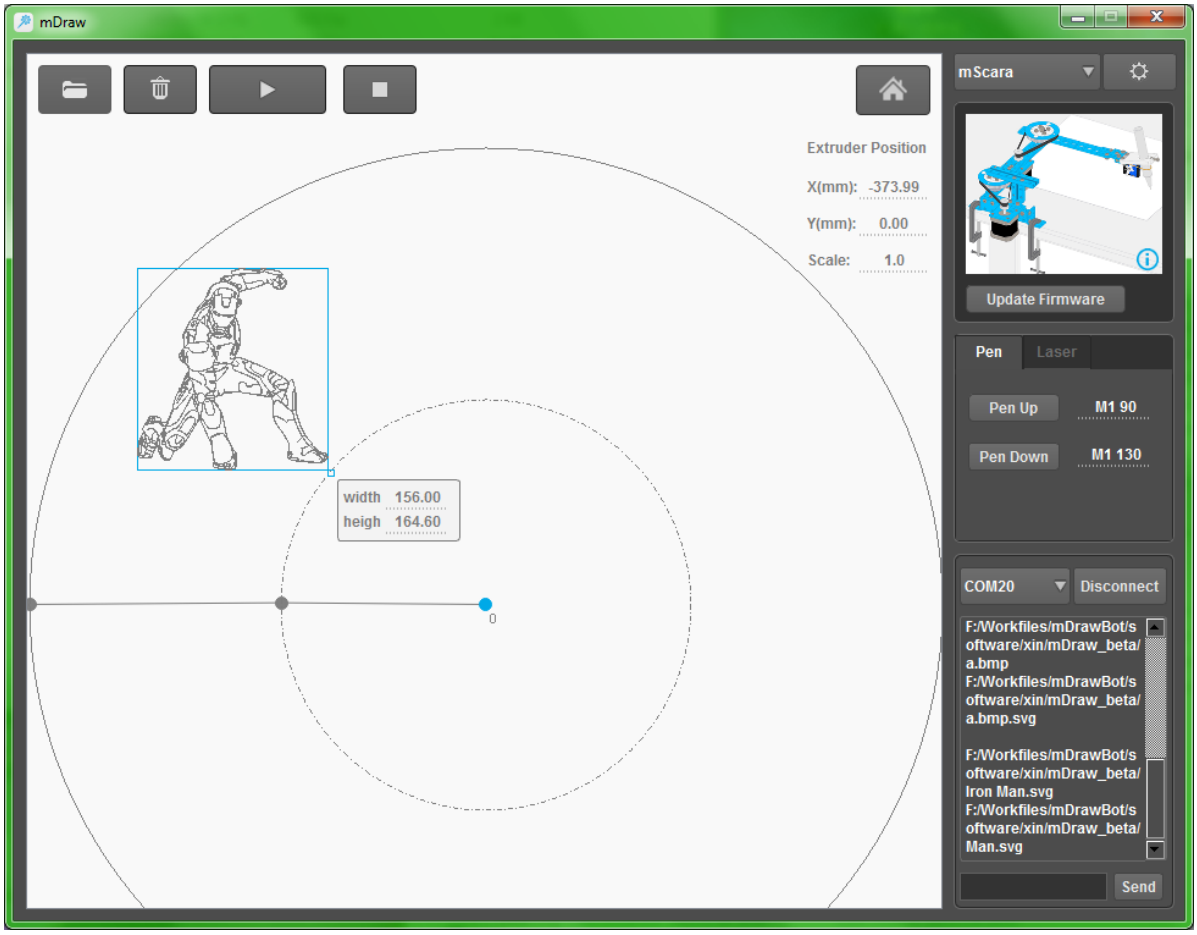
Please check the function of buttons in main UI as below figure.

Click in the Drawing zone, robot will move accordingly. Drag the loaded SVG graphics in drawing zone could adjust the size and position of graphics. Size can be adjusted by inputting values in the right-down window directly. Besides, the drawing ratio can be changed in Setting for mSpider and mCar.



Load in BMP file, mDraw will pop out a dialogue for converting SVG file. Follow the prompts, BMP file can be converted into SVG file.

Load in Iron Man SVG file, drawing is as follows.



5. Setting

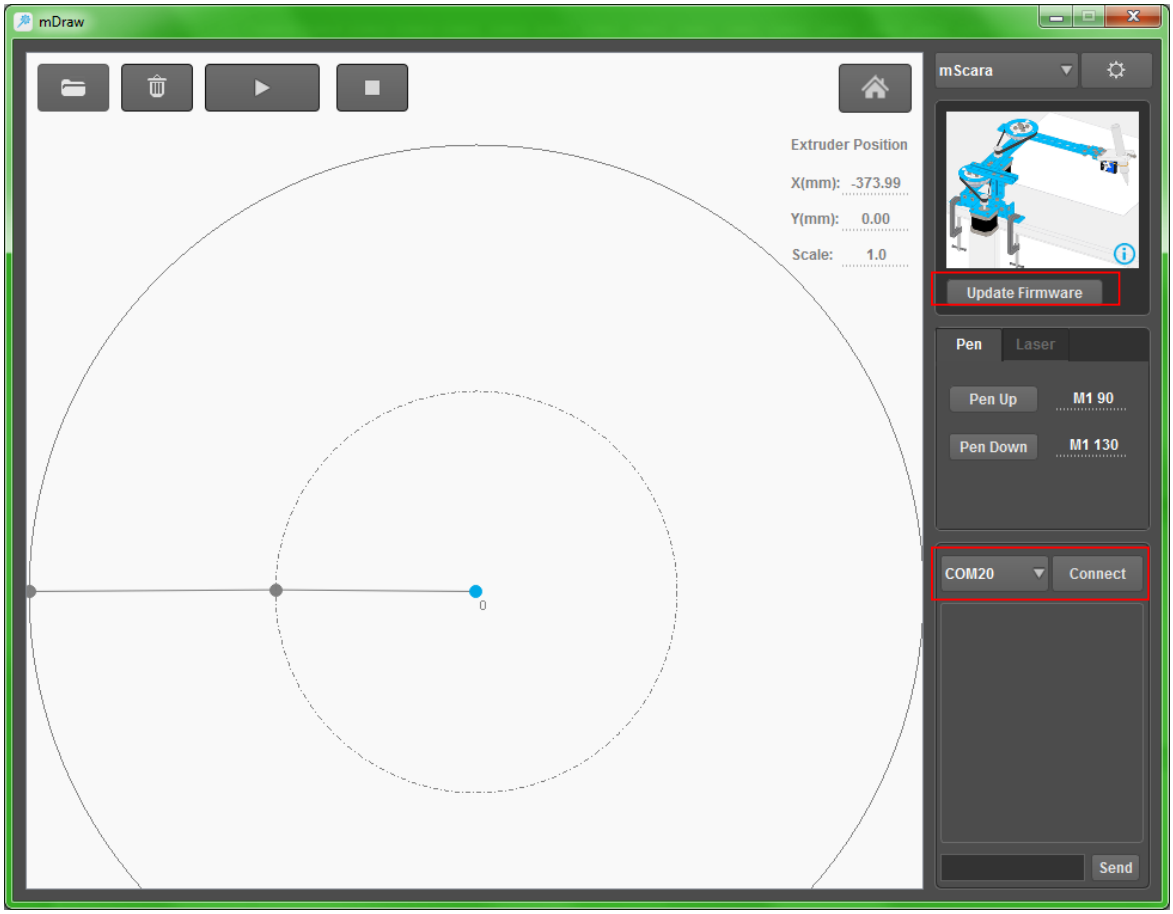
（1） mScara (Standard Version)

a. Check and make sure all wiring is correct. Connect robot to computer via USB cable.[3]

b. Select the correct COM port, click “connect”, and then click “Update Firmware”).[4]

[3] For Bluetooth connection, please be sure that you’ve installed the Bluetooth driver software in computer.

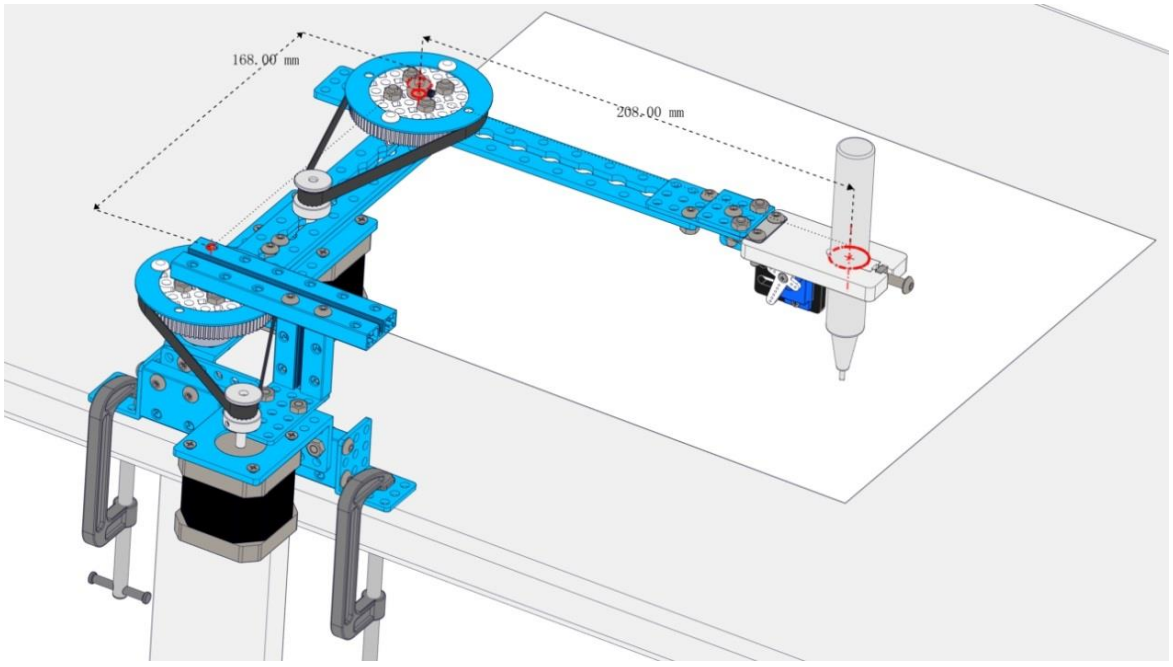
[4] You need to click “update firmware” for the first time you use a new configuration, and each time you switched robot configuration.

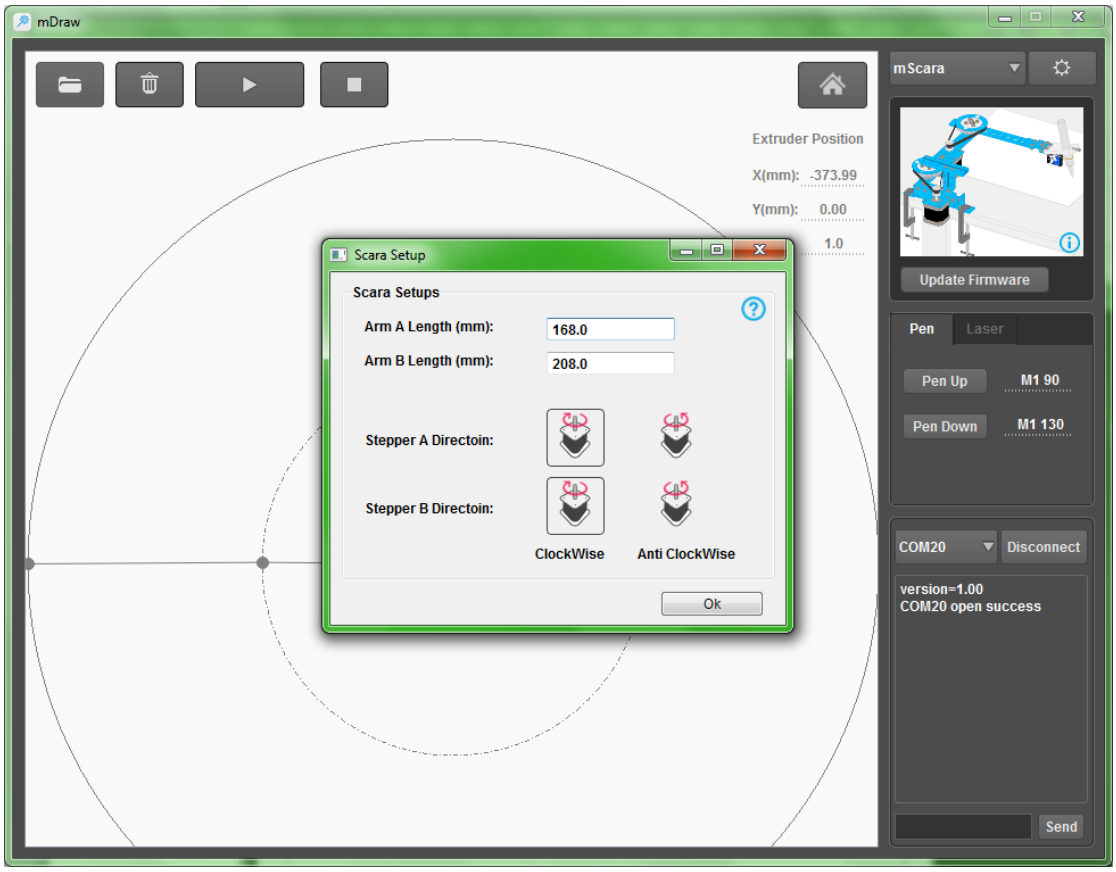


c. Click button ☼ to entry the setup window. Please keep default value unless you changed the A/B arm length of mScara. Please measure the length of A arm before you use mScara.

d. Please keep graphics stay in the circle zone, or it will cause abnormal drawing.

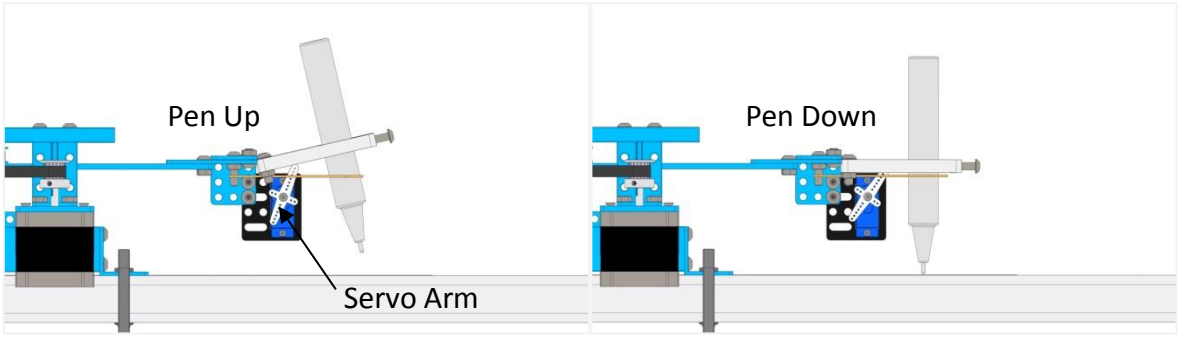
e. Please turn off the power on mainboard after settings settled. Moving mScara to specific position in below figure by hand, and then turn on the power of mainboard.

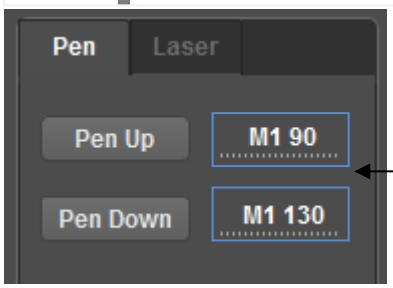




（2） Configurations of Pen Up and Pen Down

Input value of Pen Up and Pen Down in mDraw, and observe the movement of mDrawBot. If the feedback is correct as below, that means your settings are correct.[5]



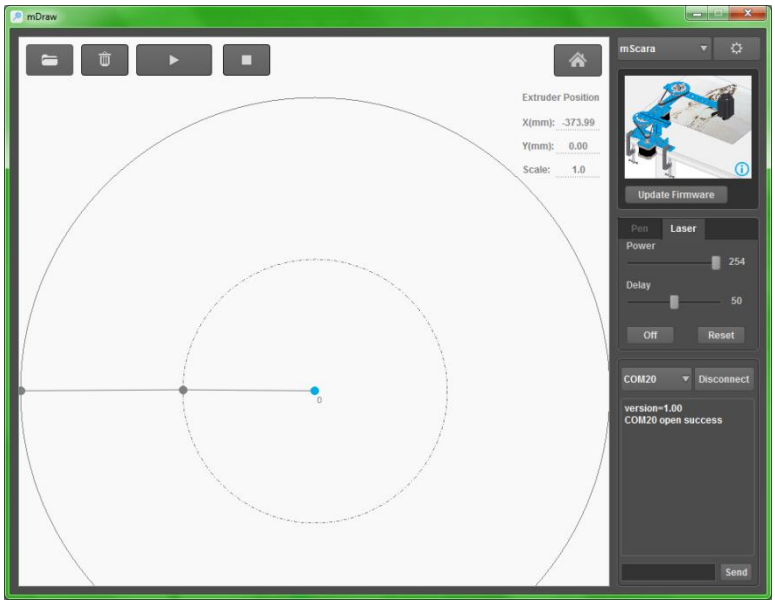


Note: the default value here is only for reference. In actual use please apply the value you got after adjusting the servo arm.

[5] If the servo arm cannot reach to below angle, please uninstall the servo arm to re-adjust the installation angle. Please adjust the tightness of rubber band if necessary.

（3） mScara (Laser Version)

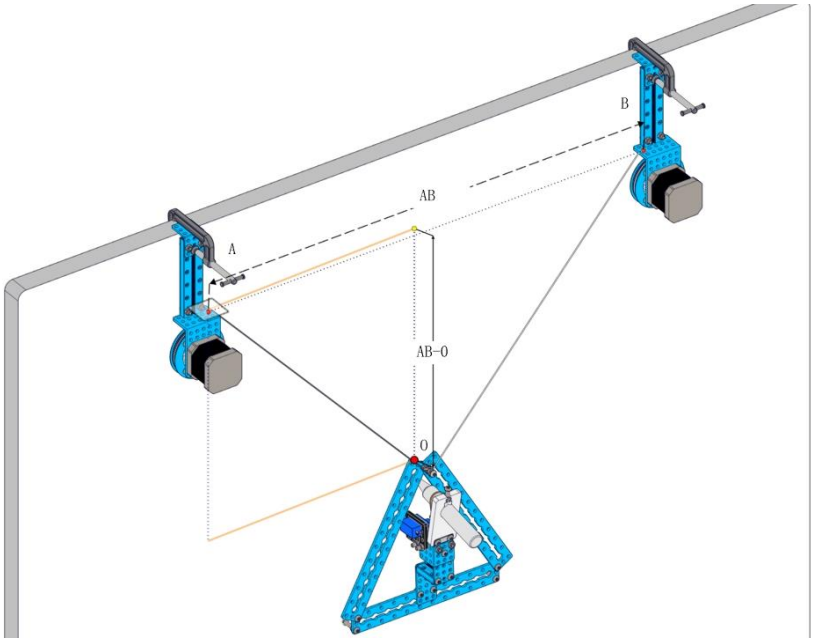
Click button “Laser” to switch mode. “Power” stands for the output laser intensity, “Delay” stands for the laser engraver time (unit: ms). Other procedures are same as mScara (Standard Version).

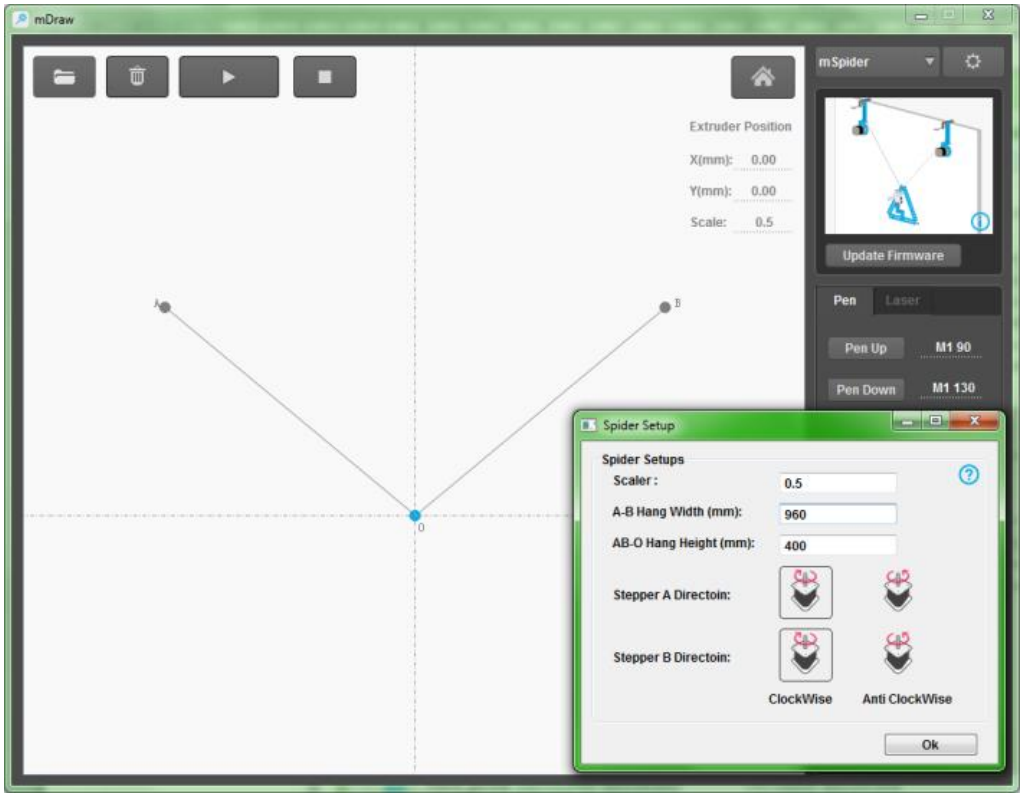


（4） mSpider

a. Check and make sure all wiring is correct. Connect robot to computer via USB cable.[6]
b. Select the correct COM port, click “connect”, and then click “Update Firmware”).[7]
c. The scaler in mSpider setup is setting the global drawing ratio, default value is 0.5 which means double the size of graphics. Below 3D figure shows what A-B Hang Width and AB-O Hang Height stand for. The point A & B are 2 holes where 2 Dacron ropes go across beam, point O is where the top screw of Pen lift mechanism touch the wall. During assembling mSpider, please measure the length of AB and AB-O, and then fill it in mDraw, click OK, done!

[6] For Bluetooth connection, please be sure that you’ve installed the Bluetooth driver software in computer.
[7] You need to click “update firmware” for the first time you use a new configuration, and each time you switched robot configuration.





d. Please check the Configurations of Pen Up and Pen Down at Here.

e. Turn on the power of mainboard, rotate **90T Timing Pulley** until the Pen lift mechanism located at the perpendicular bisector of AB. Other procedures are same as mScara (Standard Version).

（5） mEggBot

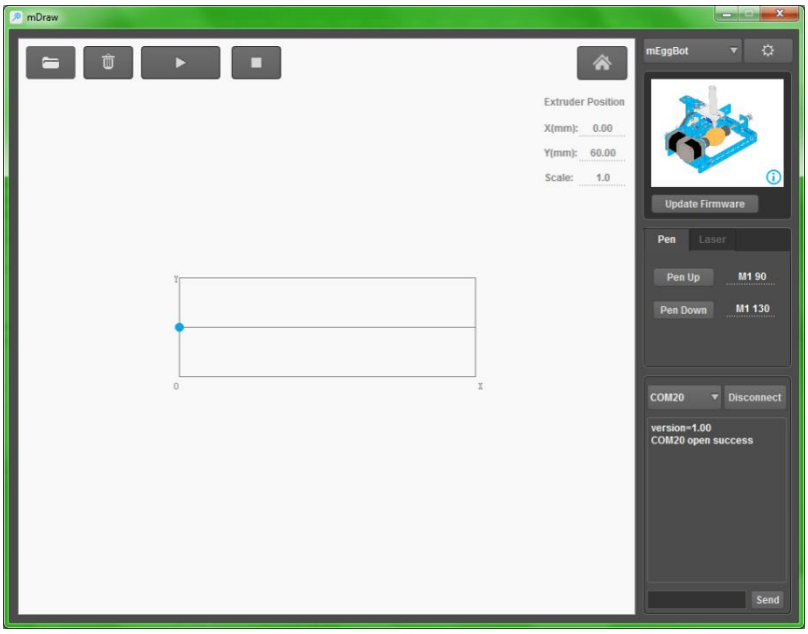
a. Check and make sure all wiring is correct. Connect robot to computer via USB cable.[8]

b. Select the correct COM port, click “connect”, and then click “Update Firmware”).[9]

[8] For Bluetooth connection, please be sure that you’ve installed the Bluetooth driver software in computer.

c. Please check the Configurations of Pen Up and Pen Down at Here. Please move the Pen lift mechanism to the center position, the object to be drew must keep clamped.

d. Other procedures are same as mScara (Standard Version).



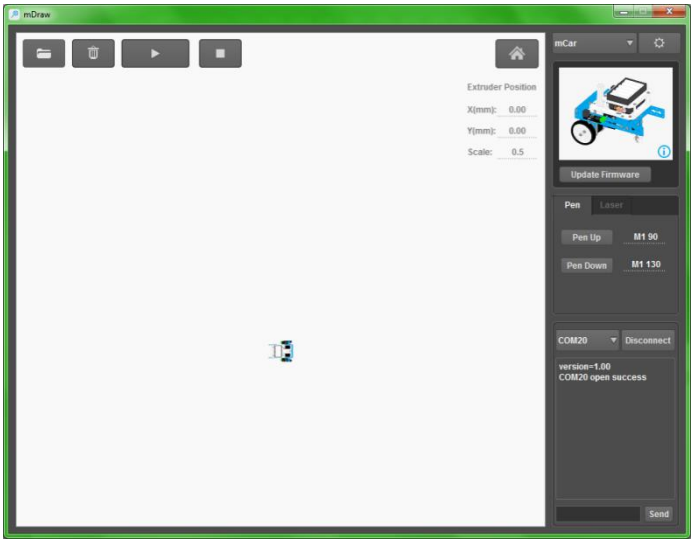
（6） mCar

a. Check and make sure all wiring is correct. Connect robot to computer via USB cable.[10]

b. Select the correct COM port, click “connect”, and then click “Update Firmware”).[11]

c. Please check the Configurations of Pen Up and Pen Down at Here.

d. mCar could possibly get slipped on floor, and influence the precision. You can upgrade it by yourself, like upgrading it to a self-balance car or a photographic car by adding a gyroscope or a camera module, and so on.



[9] You need to click “update firmware” for the first time you use a new configuration, and each time you switched robot configuration.

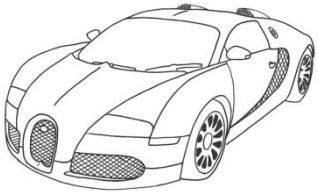
[10] For Bluetooth connection, please be sure that you’ve installed the Bluetooth driver software in computer.

[11] You need to click “update firmware” for the first time you use a new configuration, and each time you switched robot configuration.

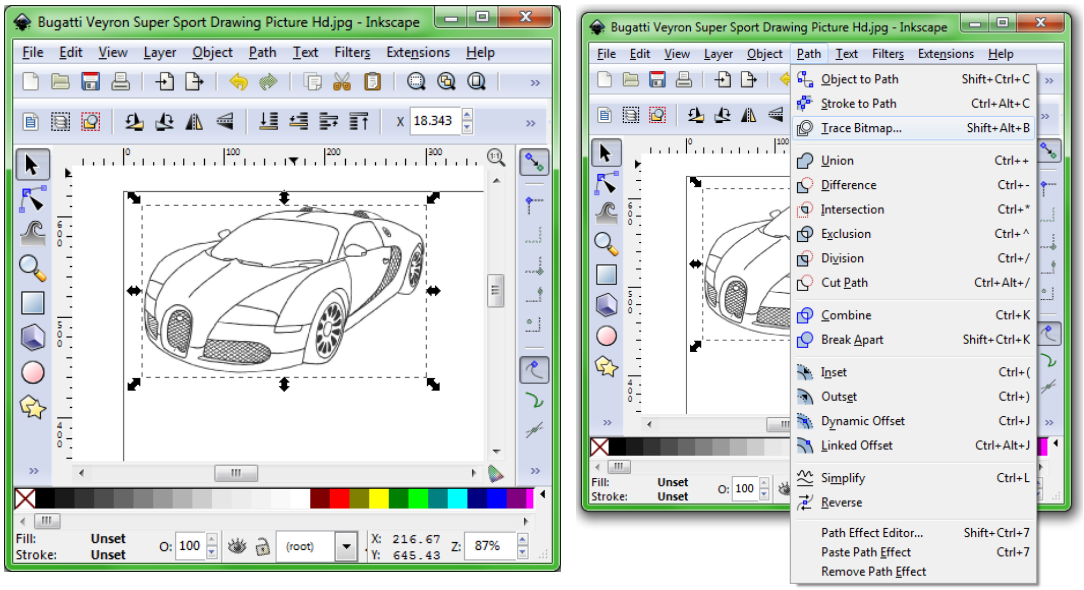
7. Samples for Software mDraw and Inkscape

mDraw supports SVG file which is based on open-source software Inkscape. For better experience of mDraw, please learn the basic operation of Inkscape.

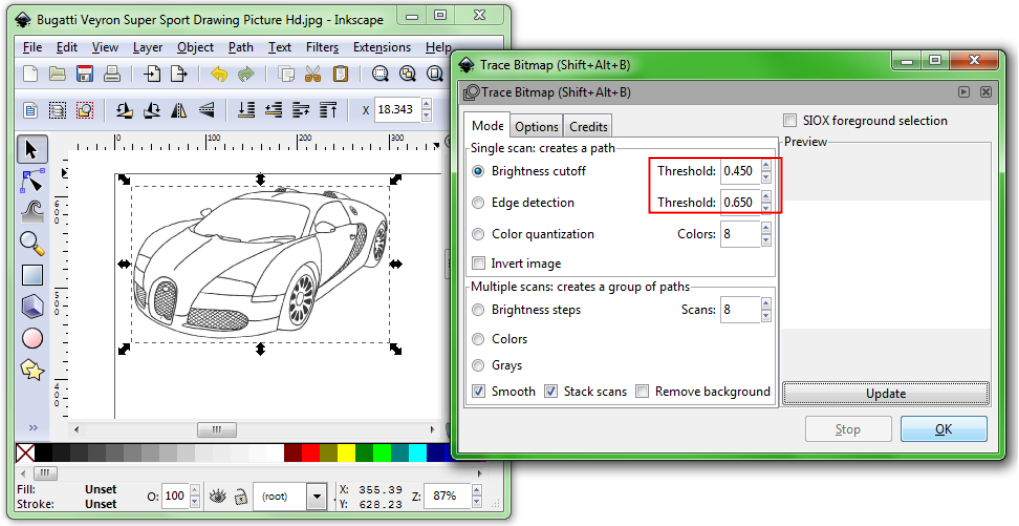
1. Install Inkscape.
2. Create a new vector graphic or open a vector graphic in Inkscape. Inkscape supports to convert other vector types into SVG file, like *.dxf, *.cdr (please save as an earlier version) files. Here you go a simple sample.
3. We suggest to choose sketch for BMP as below sample.



4. Open a BMP file with Inkscape, select the picture, and then click Path->Trace Bitmap.

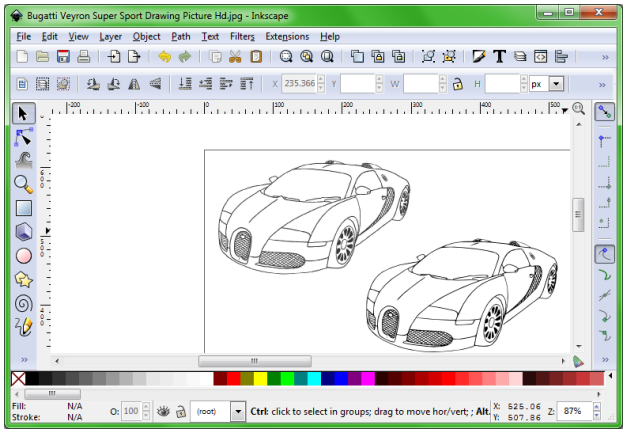


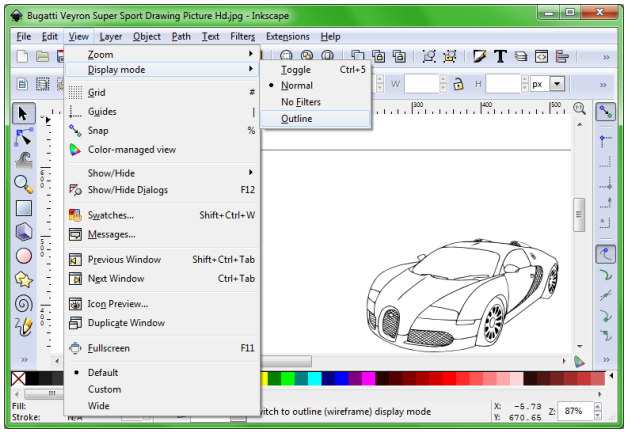
5. Set value for Threshold, click Update->OK.



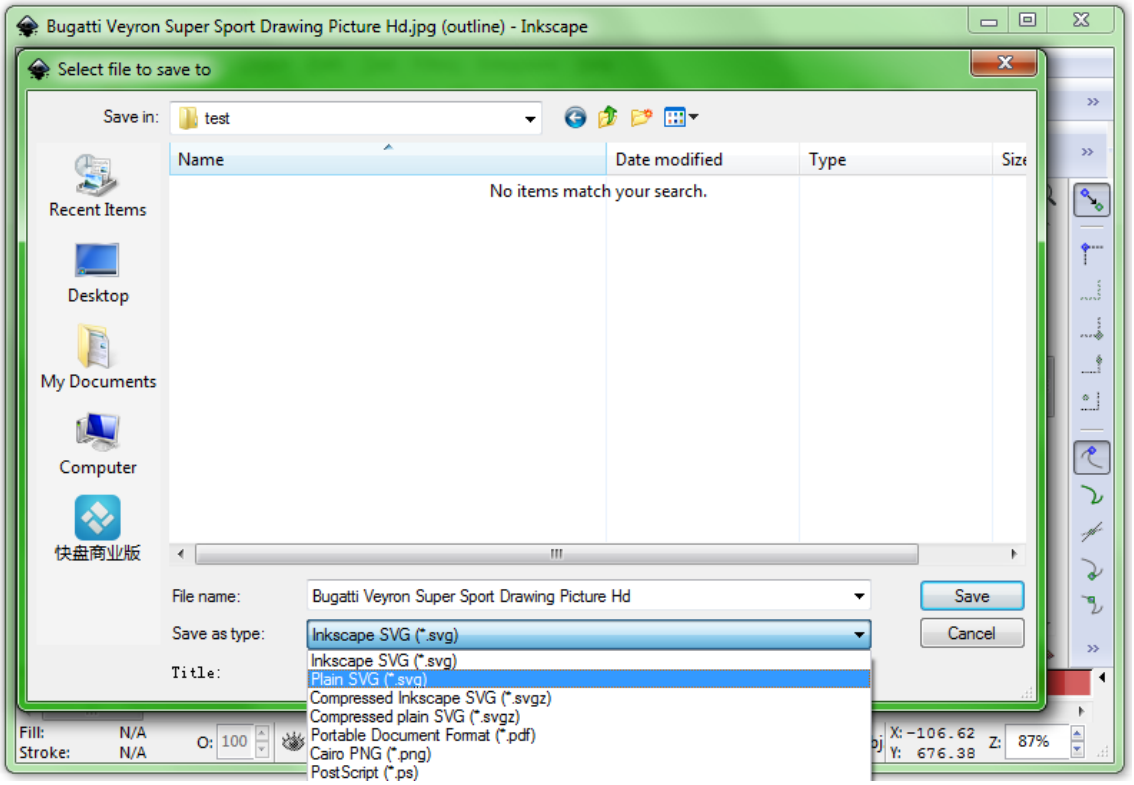
6. Drag out the generated vector graphic, select View -> Display mode -> Outline. Delete

the original BMP picture.





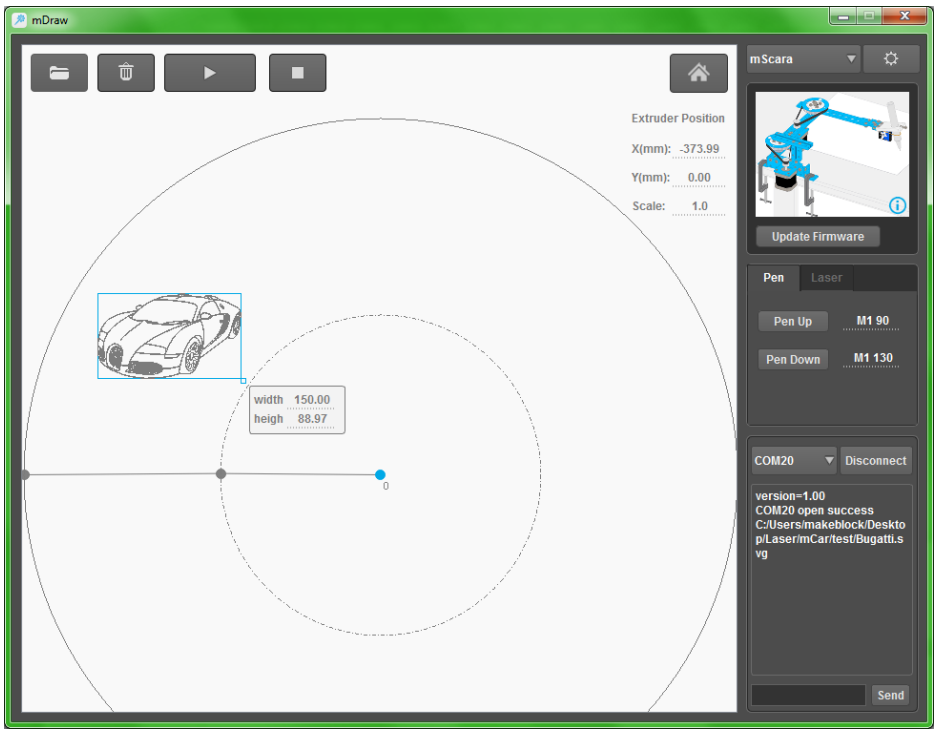
7. Save file as type Plain SVG (*svg)[12].



8. Open the SVG file you just saved as below, adjust the position and size, and then click button  to start drawing.[13]

[12] Please be sure to choose the correct file type, or it will be unsupported.

[13] Before you start drawing anything, please check the manual and be sure that the mDrawBot is located at the same position in software.


mDraw
mScara
Extruder Position
X(mm): -373.99
Y(mm): 0.00
Scale: 1.0
Update Firmware
Pen
Laser
Pen Up
M1 90
Pen Down
M1 130
width 150.00
heigh 88.97
COM20
Disconnect
version=1.00
COM20 open success
C:/Users/makeblock/Desktop/Laser/mCar/test/Bugatti.svg
Send

8. FAQs

Q1. Why cannot install driver software for Makeblock Orion?

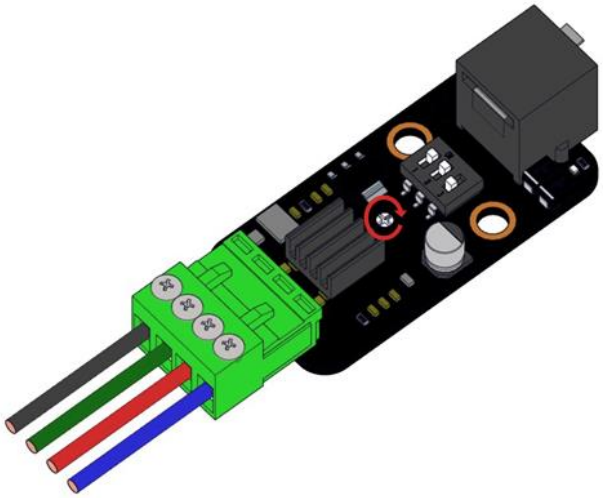
A: If your computer cannot install driver software for Makeblock Orion successfully, please download the driver installation here and install it manually.

Makeblock Orion driver download link:

http://learn.makeblock.cc/driver_installation/

Q2. How to adjust the torque and electric current of stepper motor?

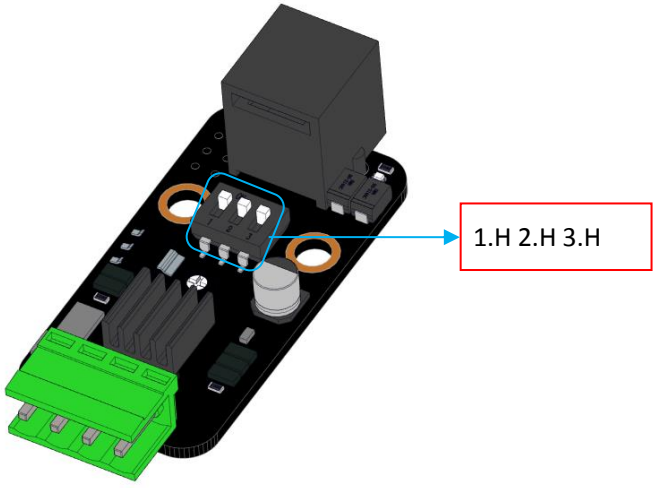
A: You can swirl the tiny knob of stepper motor driver module in "+" direction to increase the torque output if it's not enough.



Q3. Why the speed and ratio are incorrect?

A： The microstep presets of mDraw is HHH 1/16. If the speed and ratio are incorrect, please check the setting. Here you go the setting chart and default setting picture.

	1	2	3
Full speed	L	L	L
1/2	H	L	L
1/4	L	H	L
1/8	H	H	L
1/16	H	H	H



Q4. How to fix the height difference of pen lift mechanism?

A: Check the levelness of your desktop, change a better desktop if it's not good enough. If your problem still exist, please check and fix the levelness of Plate 0324-184.

Q5. How to download the related sources and 3D models?

A: Please download related sources at our website.

Q6. How to convert BMP into SVG?

A: Please refer to below teaching videos.

https://www.youtube.com/watch?v=W0V-4O9x9Uk

https://www.youtube.com/watch?v=-yBMB5KtZj4

Q7. How to share your work?

A: Welcome to upload and share you work at http://forum.makeblock.cc/category/showcase.

Q8. How to contact us?

A: Welcome to email us at support@makeblock.cc. Or you can contact us at below.

https://www.facebook.com/Makeblock?ref=br_tf

https://plus.google.com/102486511775733872783/posts

https://twitter.com/Makeblock

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